MIL-T-11503C (MU)
10 May 1963
SUPERSEDING
MIL-T-11503B (Ord)
16 May 1960

MILITARY SPECIFICATION

TRACERS
M5A2B1, M5A3B1, AND T5E3, PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the parts for, and the loading, assembling and packing of M5A2B1, M5A3B1 and T5E3 tracers for use in Artillery Ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieves; Standard, Testing.

MILITARY

JAN-B-153 - Barium, Peroxide
JAN-G-155 - Graphite, (For Use in Ammunition).
JAN-M-382 - Magnesium Powder (For Use in Ammunition).
JAN-M-454 - Magnesium-Aluminum Alloy, Powdered.
MIL-A-2550 - Ammunition and Special Weapons: General Specification for.
MIL-P-20307 - Polyvinyl Chloride (For Use in Pyrotechnics).
MIL-S-20322 - Strontium Nitrate, Non-Hygroscopic.
MIL-C-20470 - Calcium Resinate.
MIL-I-45208 - Inspection Requirements, General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

FSC: 1315

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Quality Assurance Terms and Definitions.
MIL-STD-286 - Propellant: Sampling, Inspection and Testing.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

ORDNANCE CORPS

8796866 - Tracer, M5A2B1, Assembly and Details.
8822553 - Tracer, M5A3B1, Assembly and Details.
75-17-13 - Tracer, T5E3, Loading, Assembly and Details.
7548341 - Box, Set-up, Packing, Ammunition, for Tracers M5A2B1 or M5A3B1, Assembly, Details, Packing and Marking.
7548342 - Box, Packing, Ammunition, for Tracers M5A2B1 or M5A3B1, Assembly, Details and Packing.
8800084 - Box, Packing, Ammunition, for Tracer T5E3.
8800085 - Carton, Packing, Ammunition, for Tracer T5E3.

PUBLICATIONS

ORDNANCE CORPS

IEL-7151748 - Index of Inspection Equipment Lists (for Tracer T5E3).
IEL-8796866 - Index of Inspection Equipment Lists (for Tracer M5A2B1).
IEL-8822553 - Index of Inspection Equipment Lists (for Tracer M5A3B1).

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3. REQUIREMENTS

3.1 Material.

3.1.1 Tracer composition, and igniter composition.-The material used in tracer and igniter compositions shall be in accordance with applicable specifications and in addition, shall be of the type or class specified as follows:

3.1.1.1 Strontium nitrate.-The strontium nitrate used shall be anhydrous and shall comply with the requirements of Specification MIL-S-20322, Grade B.

3.1.1.2 Magnesium-aluminum alloy, atomized or powdered.-The atomized or powdered magnesium-aluminum alloy used shall comply with the requirements of Specification JAN-M-454, Type B, except that it shall comply with the following granulation requirements using U.S. Standard Sieves complying with the requirements of Specification RR-S-366.

Through No. 100 sieve, minimum (min.) - 85 percent
Through No. 200 sieve, maximum (max.) - 15 percent

3.1.1.3 Barium peroxide.-The barium peroxide (BaO2) used shall be chemically pure, min. 88 percent BaO2 and shall comply with the requirements of Specification JAN-B-153, Class A.

3.1.1.4 Magnesium powder.-The magnesium powder used shall comply with the requirements of Specification JAN-M-382, Type III, Granulation 15.

3.1.1.5 Calcium resinate.-The calcium resinate used shall comply with the requirements of Specification MIL-C-20470, either Type I, or Type II.

3.1.1.6 Graphite.-The graphite used shall comply with the requirements of Specification JAN-G-155.

3.1.1.7 Polyvinyl chloride.-The polyvinyl chloride shall comply with the requirements of Specification MIL-P-20307.

3.1.2 Parts.-The parts shall be in accordance with the applicable drawings and specifications.

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3.2 Tracers.-The tracers shall comply with all requirements specified on Drawing (dwg.) 8796866, 8822553 or 75-17-13, as applicable, and with all the requirements specified in applicable specifications.

3.3. Composition.-The exact percentages of the ingredients indicated below, which comprise the specified composition, may be varied by the manufacturer; no ingredients other than those specified shall be utilized. (See 6.4)

3.3.1 Tracer composition.-The ingredients of the tracer composition, shall be as specified in Table I.

TABLE I

Constituent Material	Percent by Weight
Strontium nitrate, anhydrous	56.0
Magnesium-Aluminum alloy, atomized or powdered (see note 1)	37.0
Polyvinyl chloride	7.0
(Note 1: Calcium resinate (included when atomized magnesium-aluminum alloy is used.)	0.5 (added)

3.3.2 Igniter composition.-The ingredients of the igniter composition, shall be as specified in Table II.

TABLE II

Constituent Material	Percent by Weight
Barium Peroxide	80.5
Magnesium (free metallic) powder	16.5
Calcium resinate	2.0
Graphite	1.0

3.3.3 Moisture content.-The moisture content of the tracer and igniter compositions at the time of manufacture of the pellets shall not exceed 0.14 percent when tested as specified in 4.3.1 (see 6.5).

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3.4 Proving ground.-The tracers shall ignite and there shall be visible evidence of burning of the tracer composition to the target when tested as specified in 4.3.2.1.

3.5 Workmanship.-All parts shall be free from dirt, chips, grease and other foreign material. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

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e. Certificates of analysis on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions

4.2.1 Lot formation.-A lot shall consist of loaded tracers produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain:

a. Tracer parts in accordance with the same drawing and drawing revision, same specification and specification revision.

b. Tracer composition and igniter composition in accordance with same specification and specification revision from one manufacturer.

c. Tracer and igniter composition constituent materials, each from one manufacturer only.

d. Tracers of one designation only.

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4.2.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Washer, closing (see drawing 8849013 covering a detail of dwg. 8796866)

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.0 percent		
201.	Protective coating with bare spot	Visual	01001
202.	Evidence of poor workmanship (see 3.5)	Visual	01002

4.2.2.2 Housing, tracer (see drawing 8853694 covering a detail of dwg. 8796866 and 75-17-13D covering a detail of dwg. 75-17-13).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Concentricity of cavity with threads (when applicable)	Gage	02001
102.	Concentricity of cavity with small outside diameter (when applicable)	Gage	02002

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103.	Pitch diameter of thread	Gage	02003
104.	Major diameter of thread, minimum, (min.)	Gage	02004
105.	Diameter of counterbore, maximum (max.)	Gage	02005
106.	Depth of counterbore, max	Gage	02006
107.	Length of crimping lip, min	Gage	02007
108.	Crack or other damage	Visual	02008
Minor:	AQL 4.0 percent		
201.	Diameter of charge cavity	Gage	02009
202.	Depth of charge cavity	Gage	02010
203.	Total length, min	Gage	02011
204.	Outside diameter of cylindrical portion, max	Gage	02012
205.	Outside diameter at top or crimping lip	Gage	02013
206.	Profile of crimping lip	Gage	02014
207.	Distance between wrench flats	Gage	02015
208.	Length of thread, min (when applicable)	Gage	02016
209.	Distance to threads from shoulder (when applicable)	Gage	02017
210.	Burr	Visual	02018
211.	Protective coating with bare spot (when applicable)	Visual	02019
212.	Evidence of poor workmanship (see 3.5)	Visual	02020

4.2.2.3 Loading assembly, prior to crimping of closing disc (see dwgs. 8796866, 8822553 and 75-17-13, as applicable)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Charge missing or not of proper height	Visual	03001
102.	Sealing washer missing	Visual	03002
103.	Charge chipped or cracked on two or more steps	Visual	03003

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Minor:	AQL 1.0 percent		
201.	Charge chipped or cracked on one step	Visual	03004
202.	Evidence of poor workmanship (see 3.5)	Visual	03005

4.2.2.4 Tracer assembly (see dwgs. 8796866, 8822553 and 75-17-13, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Closing disc missing	Visual	04001
102.	Closing washer missing	Visual	04002
103.	Crimp incomplete	Visual	04003
104.	Closing disc damaged or loose	Visual-Manual	04004
Minor:	AQL 1.50 percent		
201.	Total length, max	Gage	04005
202.	Crimp not coated with acid proof paint (when applicable)	Visual	04006
203.	Marking misleading or unidentifiable	Visual	04007
204.	Evidence of poor workmanship (see 3.5)	Visual	04008

4.2.2.5 Box, set-up, prior to sealing (see dwg. 7548341)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 0.40 percent		
201.	Number of tracers incorrect	Visual	05001

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4.2.2.6 Box, set-up or packing carton (see dwg. 7548341 or 8800085, as applicable)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Label missing (7548341 only)...	Visual	06001
102.	Marking misleading or unidentifiable (7548341 only)..	Visual	06002
Minor:	AQL 1.50 percent		
201.	Box or carton damaged.....	Visual	06003
202.	Sealing strip incomplete.....	Visual	06004
203.	Sealing strip incompletely sealed.....	Visual	06005
204.	Contents loose.....	Manual	06006

4.2.2.7 Sealed wood packing box (see dwg. 7548342 or 8800084, as applicable)

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Box damaged.....	Visual	07001
102.	ICC nomenclature missing or unidentifiable.....	Visual	07002
103.	Strapping missing, or loose....	Visual-Manual	07003
Minor:	AQL 1.0 percent		
201.	Marking misleading or unidentifiable.....	Visual	07004
202.	Strapping improperly engaged...	Visual-Manual	07005

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4.2.3 Testing

4.2.3.1 Moisture and total volatile content of igniter composition (code no. 08001) and tracer composition (code no. 08002) Major defects, (See 3.3.3).-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test a minimum of one randomly selected sample from each batch or from each 8 hours production, whichever is the lesser amount. Composite samples shall not be used in determining compliance. Failure to comply with the applicable requirement shall be cause for rejection of the parent quantity from which the sample was drawn. Testing shall be performed as specified in 4.3.1.

4.2.3.2 Proving ground (see 3.4).-The following test shall be performed as specified in 4.3.2.1.

4.2.3.2.1 Functioning, major defect, code no. 04009.

4.2.3.2.1.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, Code letter I and an AQL of 1.50 percent. Calculations shall be in accordance with example B2, page 38.

4.2.3.2.1.2 After the three consecutive lots have met the criteria of 4.2.3.2.1.1, the sampling plan and procedures for this test shall be in accordance with Standard MIL-STD-414, Section B, Table B-3, code letter F and AQL of 1.50 percent. Calculations shall be in accordance with example B2, page 38. If any tracer fails to ignite or fails to produce visible evidence of burning to the target, the lot shall be rejected.

4.2.3.2.2 Refiring.-If for any reason the proof director considers that the tests conditions have detrimently affected the test results he shall declare the results invalid and perform a new test on additional tracers as required.

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4.2.4 Inspection Equipment.-Index of Inspection Equipment Number IEL 7151748, 8796866 or 8822553 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement of parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and wherever inspection equipment drawings are needed.

4.2.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 Government use of contractor's inspection equipment.-The contractor shall as required make available to the Government all inspection equipment necessary for determining conformance with contract requirements. Operations of the equipment and verification of its accuracy shall be supplied by the contractor, as required by the Government in performance of its examination or test.

4.3 Test methods and procedures.

4.3.1 Determination of moisture and total volatile content of tracer and igniter compositions.-The moisture and total volatile content of these compositions shall be determined in accordance with Standard MIL-STD-286, Method 101.1.

4.3.2 Proving ground.

4.3.2.1 The sample tracers, M5A2B1 and M5A3B1, shall be assembled to Shot HVTP-T, T209. The shot with assembled tracers shall be assembled to cartridge cases loaded with a charge of

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stock propellant calculated to give a pressure of approximately 45,000 pounds per square inch (psi) for the M5A2B1 and M5A3B1, and 43,500 psi., for the T5E3 tracers. The assembled rounds shall be fired from the appropriate weapon and observation made for ignition of the tracers and visible evidence of burning of the tracer composition to a 2000 meter target.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-The tracers shall be preserved and packaged in accordance with dwg. 7548341 or 8800085, as applicable.

5.2 Packing.

5.2.1 Level C.-The set-up boxes shall be packed in accordance with dwg. 7548342 or 8800084, as applicable.

5.3 Marking.-The containers shall be marked in accordance with dwgs. 7548341 and 7548342 or 8800085, as applicable.

5.4 Data cards.-Data cards shall be prepared for each lot in accordance with the information specified in Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

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6.3.1 Inspection equipment lists.-Inspection equipment lists (IEL) indicate the availability of inspection equipment design by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes indicating "commercial item" or "contractor design". Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.2 and 6.3.3. Action required by the contractor with respect to "commercial item" inspection equipment is described in paragraph 6.3.4. The contractor will be required to prepare drawings for all the equipment coded as "contractor design" in the number column. The code will further indicate whether detailed or specification control drawings are required. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 Ordnance Designs.-Ordnance designs are detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Ordnance design is listed for an item of inspection equipment. Ordnance designs fall into two basic classifications; mandatory and non-mandatory. When an IEL references mandatory Ordnance designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor prepared designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.3 Specification control drawings.-Specification control drawings depict the device in outline, descriptive, diagrammatic, or pictorial form only and specify the required performance or other characteristics. Contractors must prepare drawings of their designs in support of specification control drawings. The IEL code will indicate whether drawings must be detailed. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirement of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

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6.3.4 Commercial items.-Commercial item inspection equipment is equipment that has a universal application for a specific function. It is comprised of items commonly used by industry and government. Contractors are not required to furnish drawings of commercial item inspection equipment but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.5 IEL codes.-The inspection equipment referenced in paragraphs 6.3.1, 6.3.2, 6.3.3, and 6.3.4 will be designated in the IEL by the following codes:

CDOF - Contractor's design responsibility on Ordnance format; ORDM-608-12.
CDCF - Contractor design responsibility on contractor format.
OD - Ordnance design.
ODMU - Ordnance design, mandatory for use.
CE - Commercial equipment.
SCD - Specification control drawing.

6.3.6 Use of standard measuring equipment.-If the contractor desires to use standard measuring equipment in lieu of government or contractor designed gages, diagrammatic sketches or other descriptions of the use of such equipment shall be submitted for approval as specified in 6.3.7.

6.3.7 Submission of contractor designs.-Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

6.4 Preparation of igniter composition, and tracer composition.- It is recommended that the tracer composition be prepared as a dry mix in a Patterson-Kelly "V" type blender. The dry ingredients are weighed, placed in the blender and mixed until a uniform blend is obtained. Small rubber balls may be used in the blending operation to reduce the formation of agglomerates. The strontium nitrate is prepared for use in the tracer composition by breaking up the agglomerates, screening and drying. The igniter composition should

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be prepared using a wet mix method. It is recommended that they be blended in a Simpson type mixer using a volatile solvent, such as 95 percent ethylalcohol or acetone, as the vehicle. The calcium resinate is dissolved in sufficient solvent to make a 10 percent solution and then added to the dry ingredients previously placed in the mixer. Sufficient solvent is added to the mix to make a thick paste. The mixer is run until the mix is granular but not dry. The mixture is then dried in an oven at 80 degrees C. for approximately two hours to remove residual volatiles. If acetone is used, the mixing room should have an adequate fume exhaust system and the operators provided with suitable protection to avoid absorption of the acetone through skin contact.

6.5 Igniter and tracer compositions moisture exposure.-The igniter and tracer compositions, and the pellets formed thereof, are hygroscopic and will deteriorate rapidly when exposed to moisture. Small amounts of moisture present in, or absorbed by the compositions or pellets prior to, or after loading into tracers, will eventually result in unsatisfactory functioning. It is highly desirable that the ingredients, compositions, pellets and loaded tracers be carefully safeguarded from exposure to moisture and humid atmospheres. For assurance of a satisfactory loaded tracer, it is recommended that the blending and pelleting buildings be air conditioned in a manner which will hold the humidity of the atmosphere as low as practicable.

6.6 Shipping.-It is not intended to ship these compositions or pellets. It is intended that the compositions or pellets thereof be prepared at the plant where loading of the ammunition will be accomplished.

6.7 Combining of proving ground tests.-When the contractor for the tracer is also the contractor for the complete round or one of the loaded components thereof, the proving ground tests may be combined in order to save expense upon agreement between the procuring activity and the contractor.

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Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number 1315-A309

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ORDNANCE CORPS

ENGINEERING ORDER

FOR REGULATION PERTAINING TO THE USE OF THIS FORM SEE ORDM-4-4 MANUAL

ENGR ORDER NO. 15550-S

SHEET 1 OF 1 SHEETS

TO:

DATE: 10 May 1963

FROM: Picatinny Arsenal, Dover, New Jersey

COMMAND SYMBOL: SMUPA-ND4

YOU ARE HEREBY NOTIFIED OF THE FOLLOWING ENGINEERING ACTION, TO BE APPLIED TO PROCUREMENT OR PRODUCTION ONLY UNDER INSTRUCTIONS ISSUED BY THE CONTRACTING AGENCY.

DESCRIPTIVE DATA

MATERIAL (ORD ITEM) Tracers, M5A2B1, M5A3B1, and T5E3, Parts for and Loading, Assembling and Packing MIL-T-11503C (MU)

AUTHORITY: Ordnance Corps Order 18-55

REASON FOR ENGINEERING ACTION: To delete static burning requirement; add ballistic test provisions and update quality assurance provisions.

ENGINEERING ACTION:

Specification MIL-T-11503C (MU) dated 10 May 1963, supersedes MIL-T-11503B (Ord) dated 16 May 1960, and is hereby released for use in connection with procurement by the Army Munitions Command.

YES	NO		YES	NO	
	X	MODIFICATION TO PRIOR PRODUCTION FEASIBLE		X	DAMO RECOMMENDED
	X	MODIFICATION RECOMMENDED/REQUIRED		X	Insp. Equip. Affected
				X	Suppl Init Prod Sample Required

APPLICATION

☐ EXISTING CONTRACTS/ORDERS NOW IN PRODUCTION ☒ FUTURE CONTRACTS/ORDERS NOT YET IN EXISTENCE
☐ EXISTING CONTRACTS/ORDERS NOT YET IN PRODUCTION

CLASSIFICATION

URGENCY OF CHANGE
MANDATORY, CLASS III (CLASS I & II AFFECTS IMMO)
☐ OPTIONAL CLASS V ☐ INFORMATIONAL CLASS VI
APPLICATION

END ITEM REF. Various Complete Rounds

☐ INITIAL RELEASE

DISTRICT ACKNOWLEDGMENT

RECEIPT ACKNOWLEDGED BY:

DATE

SIG: CONTRACTING OFFICER OR AUTH. REP. ORD DIST.

AUTHORIZATION SECURED

INDICATE:
☐ OCO
☒ COMMAND

APPROVED BY:

J.W. RATHJEN
Acting Ch, Spec Eng Section
Quality Assurance Division

CLASS I - PRIOR ITEM IS UNSUITABLE FOR ISSUE
CLASS II - PRIOR ITEM IS UNSUITABLE FOR COMBAT USE
CLASS III - NON INCORPORATION MAY CREATE DIFFICULTY
CLASS IV - INCORPORATION PROVIDES IMPROVEMENT
CLASS V - INCAPABLE OF AFFECTING PART

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